HENRY R. SCHARF

CONTACT INFORMATION

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RESEARCH EMPLOYMENT

Assistant Professor *2019–Current*
San Diego State University, San Diego, California

Postdoctoral Fellow *2018–2019*
Colorado State University, Fort Collins, Colorado

Research Assistant *2014–2017*
Colorado State University, Fort Collins, Colorado
Supervisor: Mevin Hooten

Research Assistant *2013–2014*
National Renewable Energy Lab, Golden, Colorado
Supervisors: Ryan Elmore, Jay Breidt

EDUCATION

Colorado State University, Fort Collins, Colorado USA
PhD in Statistics *2017*
Advisor: Mevin Hooten

MS in Statistics *2014*

University of Arizona, Tucson, Arizona USA
MEd in Teaching and Teacher Education *2008*

BS in Mathematics and Physics *2007*
Cum laude with Honors

PUBLICATIONS

Williams, P. J., X. Lu, **H. R. Scharf**, M. B. Hooten (2023). Embracing asymmetry in nature: How to account for skewness in ecological data. *Ecological Informatics*, 75: 102085.

Boulil, Z. L.*, J. W. Durban, H. Fearnbach, T. W. Joyce, S. G. M. Leander, **H. R. Scharf** (2022). Detecting changes in dynamic social networks using multiply-labeled movement data. *Journal of Agricultural, Biological and Environmental Statistics*.

Scharf, H. R. (2022). Local indicators of spatial association (LISA). *Wiley StatsRef: Statistics Reference Online*.

Scharf, H. R., X. Lu, P. J. Williams, M. B. Hooten (2022). Constructing flexible, identifiable and interpretable statistical models for binary data. *International Statistical Review*, 90: 328–345.

Raiho, A. M., **H. R. Scharf**, C. A. Roland, D. K. Swanson, S. E. Stehn, and M. B. Hooten (2022). Searching for refuge: A framework for identifying site factors conferring resistance to climate-driven vegetation change. *Diversity and Distributions*, 28(4), 793–809.

*Student contibutor

Scharf, H. R., A. M. Raiho, S. Pugh, C. A. Roland, D. K. Swanson, S. E. Stehn, M. B. Hooten (2022). Multivariate Bayesian clustering using covariate-informed components with application to boreal vegetation sensitivity. *Biometrics*, 78: 1427–1440.

Reimer, J. R., J. Arroyo-Esquivel, J. Jiang, **H. R. Scharf**, E. M. Wolkovich, K. Zhu, C. Boettiger (2021). Noise can create or erase long transient dynamics. *Theoretical Ecology*, 14: 685–695.

Scharf, H. R. (2021). Statistical analysis of animal movement: Understanding behavior through hierarchical parametric models. *Notices of the American Mathematical Society*, 68(6), 911–924.

Scharf, H. R., F. Buderman (2020). Animal movement models for multiple individuals. *Wiley Interdisciplinary Reviews: Computational Statistics*, e1506.

Scharf, H. R., M. B. Hooten, R. R. Wilson, G. M. Durner, T. C. Atwood (2019). Accounting for phenology in the analysis of animal movement. *Biometrics*, 75: 810–820.

Hooten, M. B, **H. R. Scharf**, J. M. Morales (2019). Running on empty: Recharge dynamics from animal movement data. *Ecology Letters*, 22, 377–389.

Hooten, M. B., **H. R. Scharf**, T. J. Hefley, A. T. Pearse, M. D. Weegman (2018). Animal movement models for migratory individuals and groups. *Methods in Ecology and Evolution*, 9, 1692–1705.

Scharf, H. R., M. B. Hooten, D. S. Johnson, J. W. Durban (2018). Process convolution approaches for modeling interacting trajectories. *Environmetrics*, e2487.

Scharf, H. R., M. B. Hooten, D. S. Johnson (2017). Imputation approaches for animal movement modeling. *Journal of Agricultural, Biological and Environmental Statistics*, 22(3), 335–352.

Hefley, T. J., K. M. Broms, B. M. Brost, F. E. Buderman, S. L. Kay, **H. R. Scharf**, J. R. Tipton, P. J. Williams, and M. B. Hooten. (2017). The basis function approach to modeling autocorrelation in ecological data. *Ecology*, 98(3), 632–646.

Scharf, H. R., M. B. Hooten, B. K. Fosdick, D. S. Johnson, J. M. London, and J. W. Durban. (2016). Dynamic social networks based on movement. *Annals of Applied Statistics*, 10(4), 2182–2202.

SPONSORED RESEARCH

Factors affecting provisioning and foraging in rapidly changing landscapes — *2023–2026*
National Science Foundation, SBE.
Total Award: $329,849. Role: Co-PI.
Status: Funded

Collaborative Research: Pacific Alliance for Low-Income Inclusion in Statistics & Data Science — *2022–2028*
National Science Foundation, S-STEM.
Total Award: $5,000,000. Role: Co-PI.
Status: Funded

Visualizing trajectory uncertainty — 2021–2022
University Grants Program, SDSU.
Total Award: $10,000. Role: PI.
Status: Funded

Broadening the impact and accessibility of animal movement models — 2020–2021
University Grants Program, SDSU.
Total Award: $10,000. Role: PI.
Status: Funded

INVITED CONFERENCE PRESENTATIONS

Detecting changes in dynamic social networks using unlabeled movement data — 2022
Joint Statistical Meetings, Washington D.C., USA

Detecting changes in dynamic social networks based on unlabeled movement data — 2021
CMStatistics, London, UK

Multivariate Bayesian clustering using covariate-informed components with application to boreal vegetation sensitivity — 2021
Joint Statistical Meetings, Seattle, WA, USA

Statistical models for dependent trajectories with application to animal movement — 2019
Joint Statistical Meetings, Denver, CO, USA
Topic contributed session for ISBA Savage Award finalists

Animal movement models for migratory individuals and groups — 2018
The Wildlife Society Annual Conference, Cleveland, OH
Symposium: *Animal movement: Advances in movement modeling and their applications*

Imputation approaches for animal movement modeling — 2018
Joint Statistical Meetings, Vancouver, BC, Canada
Invited session

Dynamic Social Networks Based on Movement — 2016
Joint Statistical Meetings, Chicago, IL
Student paper award winner (ENVR)

DEPARTMENTAL SEMINARS

Detecting changes in dynamic social networks based on unlabeled movement data — 2021
Department of Mathematics and Statistics, California State University Long Beach

Movement data reveal dynamic social relationships — 2021
Department of Statistics, University of British Columbia

Movement data reveal dynamic social relationships — 2021
Department of Statistics, Kansas State University

Multi-layered convolutional Gaussian process models for animal movement — 2020
Computational Sciences Research Center, San Diego State University

Statistical models for heterogeneous animal movement — *2019*
Department of Statistics & Data Science, University of Texas at Austin

CONTRIBUTED PRESENTATIONS

Identifying species-level vegetation cover using Sentinel-2 imagery — *2022*
ASA Section on Statistics and the Environment Workshop, Provo, Utah

Accounting for phenology in the analysis of animal movement — *2018*
International Statistical Ecology Conference, University of St. Andrews, Scotland

Process convolution approaches for modeling interacting trajectories — *2017*
Joint Statistical Meetings, Baltimore, MD

Process convolution approaches for modeling interacting trajectories — *2017*
Statistics Department at Colorado State University, Fort Collins, CO.
Poster

Dynamic Social Networks Based on Movement — *2016*
International Statistical Ecology Conference, Seattle, WA

Spatiotemporal Models for Animal Social Structure — *2015*
Joint Statistical Meetings, Seattle, WA

Spatiotemporal Models for Animal Social Structure — *2015*
Statistical and Applied Mathematical Sciences Institute
ECOL: Transitional Workshop Research Triangle Park, NC

Dynamic social networks based on movement — *2015*
Graduate Student Showcase, Fort Collins, CO.
Poster

Spatiotemporal models for animal social structure — *2015*
Statistics Department at Colorado State University, Fort Collins, CO.
Poster

Novel visualization and analysis for extreme-scale wind turbine simulations — *2014*
Conference on Data Analysis, Santa Fe, NM.
Poster

TEACHING

R Programming and Data Science (SDSU: STAT 410) — *Fall 2022*
Instructor

Bayesian Statistics (SDSU: STAT 676) — *Spring 2022*
Instructor

Applied Spatio-Temporal Statistics (SDSU: STAT 596/696) — *Spring 2021*, *Spring 2020*
Instructor

Statistical Computing (SDSU: STAT 580) — *Fall 2022*, *Fall 2021*, *Fall 2020*
Instructor

Introduction to Statistical Learning (SDSU: STAT 596) — *Fall 2019*
Instructor

Statistics for Business Students (CSU: STAT 204) — *Spring 2013*
Instructor

Statistics for Business Students (CSU: STAT 204) — *Fall 2012*
Teaching Assistant

Geometry and intermediate algebra — *2009–2010*
Teacher
Tucson High Magnet School, Tucson, Arizona

Intermediate algebra — *2009*
Teacher
Flowing Wells High School, Tucson, Arizona

ADVISING

Masters students

Angelica Rivera	*current*
Hugo Rosales Portillo	*current*
Navid Nezamabadi	*Summer 2022*
Jonathan Schierbaum	*Spring 2022*
Zaineb Boulil	*Spring 2021*
Jennifer Betancourt	*Spring 2021*
Kristine Dinh	*Spring 2021*

Undergraduate students

Elena Dubocanin	*Winter 2022*

WORKSHOPS & SHORT COURSES

Workshop: Statistical Models for Animal Movement
Co-instructor
Day-long workshop on animal movement modeling with examples in R

International Association for Landscape Ecology annual meeting — *2019*
Fort Collins, CO, USA

International Statistical Ecology Conference — *2018*
University of St. Andrews, Scotland

Training in Bayesian Modeling for Practicing Ecologists. — *2016*
Co-instructor
NSF-supported two-week workshop
Intensive training in Bayesian modeling
Colorado State University, Fort Collins, CO

R Workshop — *2015*
Co-instructor
Day-long tutorial on R for ecologists
Colorado State University, Fort Collins, CO

Tutorial on Parallel Programming in R — *2015*
Co-Developer
Workshop on Parallel Computing, UC Boulder/CSU

Tutorial on Parallel Programming in R — *2015*
Co-Developer
Conference on Statistical Practice 2015, New Orleans, LA

AWARDS

ISBA Savage Award - Applied Methodology *2018*
The Savage Award, named in honor of Leonard J. "Jimmie" Savage, is bestowed each year to two outstanding doctoral dissertations in Bayesian econometrics and statistics.
Amount: $750 + $500 travel support

James L., M. Leslie, & Edna Madison Memorial Award *2017*
This award is given each year to the student selected by the statistics faculty at Colorado State University as the outstanding graduate student in the Department of Statistics and/or the Department of Mathematics
Amount: $675

J.E. Moyal Award for Applied Probability *2016*
This award, which is not given annually, is given in recognition of academic excellence in applied probability by the statistics faculty at Colorado State University.
Amount: $500

Student paper award: Section on Statistics and the Environment *2016*
This award includes travel assistance to present in an invited session at the Joint Statistical Meetings 2016 in Chicago, IL.
Amount: $1000

Joint Statistical Meetings travel award from the ASA *2014*
Amount: $900

SAMSI Program on Mathematical and Statistical Ecology opening workshop travel award *2014*
Amount: $470 + lodging

Pan-American Advanced Study Institute travel award from STATMOS *2014*
Amount: $4000

Graybill Award for Excellence in Linear Models *2014*
This award was established in 1975 to honor the many contributions of Frank Graybill, founder and former chairman of the Statistics Department at Colorado State University. It is awarded annually to a graduate student who is selected as the top student in linear models.
Amount: $1000

James S. Williams Scholarship *2013*
This scholarship was established in 1991 in memory of Professor James S. Williams to encourage new students to join the graduate program in Statistics at Colorado State University.
Amount: $1,500

Design and Analysis of Experiments Conference travel award *2012*
Amount: $700

SERVICE

Award committee for ISBA Savage Award - Applied Methodology for outstanding dissertations in Bayesian econometrics and statistics *2021*

Secretary: American Statistical Association Section on Statistics and the Environment (ENVR) *2020*

Treasurer: American Statistical Association Section on Statistics and the Environment (ENVR) *2019*

Associate Editor:
Data Science in Science *2021–Current*

Reviewer for the following journals:
Annals of Applied Statistics
Computational Statistics and Data Analysis
Environmetrics
Environmental and Ecological Statistics
Journal of Agricultural, Biological, and Environmental Statistics
Journal of Computational and Graphical Statistics
Ecological Monographs
Ecology
Methods in Ecology and Evolution
Network Science
Proceedings of the Royal Society: Series B
PLOS ONE

R PACKAGES

Scharf, H. R., K. Dinh*, H. Rosales*, A. Rivera* (2023). anipaths: Animation of observed trajectories using spline- or state-space model-based interpolation. R package version 0.10.2.

COMMITMENT TO DIVERSITY

Panelist: 2019 Career Issues Panel Women in Science Symposium *2019*

Panel moderator, organizer: Women in Science Symposium *2018*
Topics of discussion included:
Practical suggestions for recruiting, retaining, and promoting women
Insight into implicit and explicit bias
Tips to initiate your own Career Issues small group
Networking opportunities with men and women from academic, industry and the Front Range community to promote gender equality

Career issues *2015–Current*
Graduate student-initiated discussion group made up of students, and junior, mid-career, and senior faculty from multiple departments. Topics included: strategies for career advancement for women, awareness of systemic sexism in academia, and increasing participation of women in academia and STEM fields in particular.

OTHER CONFERENCES & WORKSHOPS

Participant in NIMBioS Investigative Workshop *Transients in Biological Systems*. *2019*

Member of the working group *Network Models* in the SAMSI 2014-15 Program on Mathematical and Statistical Ecology (ECOL). *2014–2015*

Pan-American Advanced Studies Institute (Buzios, Brazil) *2014*
Workshop focused on spatiotemporal statistics, with a special emphasis on international collaboration.

*Student contibutor

	Design and Analysis of Experiments Conference	*2012*
PROFESSIONAL AFFILIATIONS	International Biometric Society WNAR region	*2017–Current*
	International Society for Bayesian Analysis	*2017–Current*
	Institute of Mathematical Statistics	*2015–Current*
	American Statistical Association ENVR section	*2013–Current*